

Unit 8 | Chapter 11: The "Cloud" (Git, GitHub & Cloud Concepts)

Learning Targets:	
❖ Technical: I can differentiate between Git (local version control) and GitHub (cloud hosting) and perform core lifecycle actions (init, add, commit, push, pull, clone). (NV 5.6.3) ❖ Technical: I can identify and compare cloud service models (IaaS, PaaS, SaaS) using industry-standard analogies. (NV 5.6.1) ❖ Technical: I can explain cloud infrastructure concepts like Scalability, Elasticity, and Deployment Models (Public, Private, Hybrid). (NV 5.6.2) ❖ Employability: I can demonstrate analytical reasoning and cognitive flexibility to independently resolve terminal errors and Git merge conflicts. (WRS 1.2.3) ❖ Employability: I can demonstrate diligence, probity, and inclusive collaboration by creating professional technical documentation (README) and managing shared repositories responsibly. (WRS 1.1.1, 1.1.2, 1.1.5)	
Compelling Question:	
How does "The Cloud" move code from a single developer's hard drive to a global audience, and why is version control the "time machine" of modern engineering?	
Score	Performance Indicators
4.0	I can: ☐ Manage a multi-developer workflow by successfully resolving Merge Conflicts and maintaining a definitive Main branch, demonstrating advanced analytical reasoning and systematic resolution. (NV 5.6.3, WRS 1.2.3) ☐ Architect a deployment strategy that uses a Hybrid Cloud model to balance cost-efficiency with data security, demonstrating strict probity with sensitive information. (NV 5.6.2, WRS 1.1.2) ☐ Evaluate a viral traffic spike scenario and determine how Elasticity prevents financial loss, displaying profound cognitive flexibility. (NV 5.6.1) ☐ Create professional project documentation (README) that promotes an egalitarian open-source environment and inclusive collaboration. (NV 5.6.3, WRS 1.1.5)
3.5	In addition to score 3.0 performance, partial success at score 4.0 content.
3.0	I can: ☐ Execute the full Git lifecycle: Initialize a repo, stage changes (add), make a snapshot (commit), and push the history to a remote repository. (NV 5.6.3) 

	☐ Identify the correct cloud service model (IaaS, PaaS, SaaS) for a given project utilizing the "Pizza as a Service" analogy. (NV 5.6.1) ☐ Perform a Git pull to update a local environment with a teammate's changes, demonstrating inclusive collaboration. (NV 5.6.3, WRS 1.1.5) ☐ Navigate the file system using CLI commands (cd, ls, pwd) with diligence and professional decorum. (NV 5.6.3, WRS 1.1.1) ☐ Demonstrate intrinsic motivation and punctuality by completing daily terminal labs on schedule. (WRS 1.1.1)
2.5	No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.
2.0	I can ☐ Define common terminology and identify syntax: ☐ Git vs. GitHub, Repository (Repo), CLI, VCS, AWS/Azure/GCP, IaaS/PaaS/SaaS, Scalability vs. Elasticity, Deploy, Merge, pwd, ls, cd, and the "Main" Branch. (NV 5.6.1, 5.6.2, 5.6.3) ☐ Identify the importance of punctuality and diligence (WRS 1.1.1), probity in securing code (WRS 1.1.2), professional decorum in commit messages (WRS 1.1.4), inclusive collaboration on teams (WRS 1.1.5), and applying analytical reasoning to terminal errors (WRS 1.2.3).
1.5	Partial success at score 2.0 content, and major errors or omissions at 3.0.
1.0	With help, partial success at score 2.0 and score 3.0 content.
0.5	With help, partial success at score 2.0 but not at score 3.0 content.
0.0	Even with help, no success or attempt made.